BIRZEIT UNIVERSITY

Electrical and Computer Engineering Department

Machine Learning and Data Science - ENCS5341

Assignment #3

Prepared by: Tasneem Shelleh 1220439

Rawand Radi 1221054

Instructor: Dr. Yazan Abu Farha

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1. Introduction

1.1. The Learning Task

The chosen task is weather prediction from travel images. Given a photograph from the collected travel-destination dataset, the objective is to classify the weather condition shown in the image into one of several predefined classes.

This task is challenging because weather is represented through subtle visual cues such as lighting, sky appearance, and overall atmospheric conditions. Unlike object recognition, where models focus on distinct objects, weather classification requires learning global scene characteristics rather than localized features.

1.1. Models Explored

To explor this classification we developed and evaluate three different models

1.2.1. K-Nearest Neighbors (KNN) Baseline

We established a baseline using a distance-based k-nearest neighbors approach with (k=1) and (k=3), based on feature vectors extracted from the images. This simple, non-parametric method serves as a reference point for evaluating the performance of more advanced models.

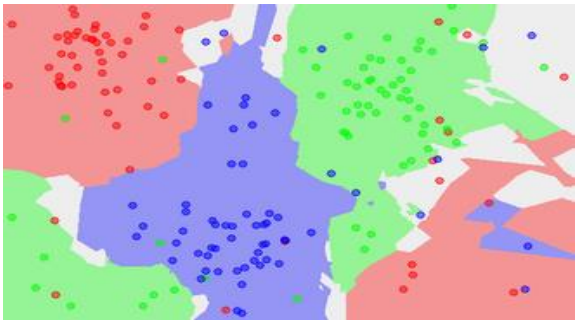

Figure 1-1: K-Nearest Neighbors (KNN) general

1.2.2. Random Forest Classifier

The first model explored is the Random Forest classifier, an ensemble learning method that builds multiple decision trees during training. Random Forests are well suited for this task because they can handle high-dimensional feature spaces, are relatively robust to overfitting, and are capable of capturing complex non-linear relationships in image features.

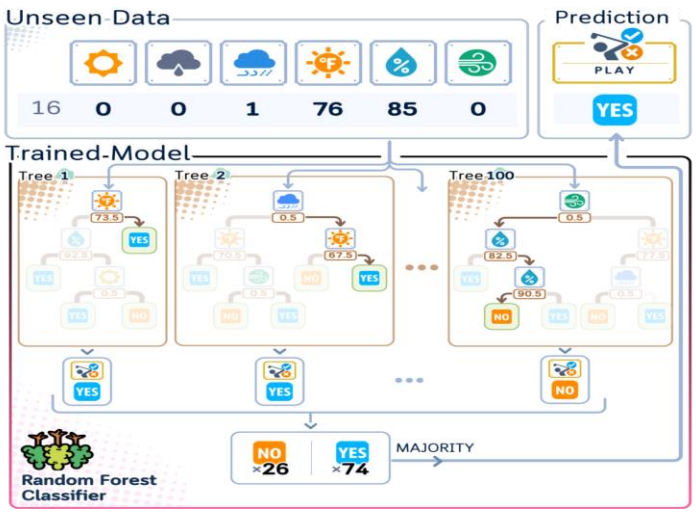

Figure 1-2: Random Forest Classifier

1.2.3 Support Vector Machine (SVM)

The second model is Support Vector Machine (SVM) classifier to predict weather conditions from travel images. The model applies a radial basis function (RBF) kernel to capture non-linear decision boundaries and is trained on the fused HSV color and SIFT texture features. Thanks to its strong generalization ability and robustness when dealing with high-dimensional data, the SVM offers an effective and reliable solution for image-based weather classification.

For feature extraction, we used a combination of HSV (Hue–Saturation–Value) color histograms and basic statistical features. The HSV color space is particularly suitable for weather classification because it separates color information from intensity, making it more robust to lighting variations compared to RGB representations.

1.3 Evaluation Metrics

To evaluate the performance of each model, accuracy was used as the primary evaluation metric, measuring the proportion of correctly classified images and providing an overall assessment of model performance.

In addition, precision, recall, and F1-score were computed for each class. These metrics offer more detailed insight into model behavior across different weather categories and are particularly important given the presence of class imbalance in the dataset.

A confusion matrix was also generated to provide a detailed comparison between predicted and true labels, allowing for analysis of misclassification patterns and identification of weather classes that are most frequently confused.

2. Data Preprocessing and Exploratory Data Analysis (EDA)

2.1 Data Preprocessing

The dataset was collected from multiple sources and stored in several CSV files, so the first preprocessing step was to merge these files into a single dataset. During the initial merge, many conflicts in column names were observed, which made the merging process inconsistent. To address this issue, the column names were first standardized by removing special characters, extra spaces, and encoding artifacts, and by converting all names to a consistent format. In addition, different variations of the same column name were mapped to unified standard column names, ensuring a clean and consistent merged dataset.

```python
COLUMN_MAPPING = {
    'image_url': ['image url', 'image_url', 'imageurl', 'url', 'image', 'picture', 'link','image_url_description'],
    'description': ['description', 'describtion'],
    'country': ['country', 'destination name','destination_name'],
    'weather': ['weather'],
    'time_of_day': ['time of day', 'time_of_day', 'timeofday'],
    'season': ['season'],
    'activity': ['activity'],
    'mood': ['mood/emotion', 'mood_emotion', 'mood', 'emotion', 'mode','moodemotion']
}
```

Figure 2-1: mapped to standard column names

After merging the files, duplicate records and any erroneous columns that appeared as a side effect of the merging process were removed. Several additional data cleaning steps were then applied to improve data quality. Invalid image URLs were filtered out by verifying that the links followed a correct URL format and pointed to actual image content. Images that could not be successfully downloaded or opened were also excluded, ensuring that all remaining samples were usable for feature extraction.

Other attributes such as descriptions, weather, mood, season, time of day, and activity, which are all text-based features, were cleaned by removing encoding artifacts, extra whitespace, punctuation, and inconsistent capitalization. All text values were normalized to a consistent format to reduce noise caused by spelling variations and formatting inconsistencies.

```python
df["description"] = (
    df["description"]
    .fillna("")
    .astype(str)
    .str.replace("\ufeff", "", regex=False)
    .str.replace(r"\s+", " ", regex=True)
    .str.strip()
)
```

Python

```python
df.sample(5)
```

Python

	image_url	description	country	weather	time_of_day	season	activity	mood
91	https://www.ricksteves.com/watch-read-listen/r...	Swiss cows grazing in the Berner Oberland Alps...	Switzerland	Sunny	Afternoon	Summer	Viewing nature / countryside	Relaxing
876	https://cdn-imgix.headout.com/media/images/ab8...	CappadociaâÃ¢s valleys filled with hot-air bal...	TÃ¼rkiye	Sunny	Morning	Spring	Hot-air balloon ride	Adventure
1176	https://i.postimg.cc/m2t5SJ2V/image3.png	A rugged mountainous landscape in Afghanistan,...	Afghanistan	Sunny	Afternoon	Summer	Hiking	Adventure
197	https://www.fairflight.de/blog/wp-content/uplo...	A bright coastal view of Waikiki Beach with tu...	USA	Sunny	Morning	Summer	Relaxing	Happiness
451	https://upload.wikimedia.org/wikipedia/commons...	The red desert dunes of Djanet, Algeria a st...	Algeria	Sunny	Afternoon	Not Clear	Exploring	Adventure

Figure 2-2: text-based features cleaning example

Categorical attributes were standardized using rule-based normalization. Weather labels were mapped into a fixed set of classes (Sunny, Cloudy, Rainy, Snowy, and Not Clear), and ambiguous or invalid entries were removed. Similarly, time of day values were normalized into consistent categories (Morning, Afternoon, Evening, and Not Clear). Mood and activity labels were consolidated into higher-level categories to reduce sparsity and improve label consistency.

Missing or invalid entries were handled either by filling them with a standard value such as Unknown when the feature was not used in the model, or by removing the corresponding records when the feature was required for training. After completing these steps, the dataset was saved in a cleaned final form and used for exploratory data analysis and subsequent model training.

After merging all CSV files, the dataset initially contained 1,457 samples with 8 features. After cleaning, the dataset was reduced to 959 samples, mainly due to removing invalid image URLs, unreadable images, and records with missing essential information. This reduction improved the overall quality of the data used for analysis and model training.

2.2 Exploratory Data Analysis

EDA was performed to better understand the structure and content of the travel dataset before building any predictive models. A combination of quantitative summaries and visual visualizations was used to highlight potential challenges, and evaluate correlations among variables.

2.3 Key Steps of the EDA

1. Initial Dataset Inspection: The dataset was loaded and initially explored by displaying sample records, checking its dimensions, reviewing data types, and verifying the absence of missing values.

2. Correlation Analysis of Labels: All categorical attributes were converted into numerical form and used to compute a correlation matrix. The resulting matrix was visualized using a heatmap.

3. Quantitative Categorical Statistics: For each label, multiple quantitative measures were computed, including: Mode (most frequent class that dominates the dataset), Entropy(label

diversity), Gini Index(class distribution across the dataset), Imbalance Ratio(severity of class imbalance), Class Proportions percentages.

4. Visualization of Weather Distribution: A bar chart was generated to display the frequency of each weather class, providing a clear visual overview of the dominant and underrepresented weather conditions in the dataset.

5. Detection of Rare Weather Classes: The percentage distribution of weather labels was calculated, and weather classes representing less than 5% of the dataset were identified.

6. Cross-Attribute Relationship Analysis: Stacked bar charts were created using cross-tabulations to analyze how weather is related to other labels.

2.4 Descriptive Statistics of the Dataset

The travel dataset consists of 959 samples and 8 categorical attributes describing image-based travel scenes. All attributes contain complete records with no missing values.
Quantitative Categorical Statistics for each label:

```
--- WEATHER ---
Mode: Sunny
Entropy: 1.172
Gini Index: 0.603
Imbalance Ratio: 26.33
Class Proportions (%):
weather
Sunny        57.66
Cloudy       21.69
Not Clear     9.38
Snowy         9.07
Rainy         2.19
Name: proportion, dtype: float64

--- SEASON ---
Mode: summer
Entropy: 1.479
Gini Index: 0.74
Imbalance Ratio: 3.6
Class Proportions (%):
season
summer       41.29
spring       19.40
winter       15.22
fall         12.62
not clear    11.47
Name: proportion, dtype: float64
```

```
--- TIME_OF_DAY ---
Mode: Afternoon
Entropy: 1.104
Gini Index: 0.641
Imbalance Ratio: 38.17
Class Proportions (%):
time_of_day
Afternoon    47.76
Evening      27.11
Morning      23.88
Not Clear     1.25
Name: proportion, dtype: float64

--- ACTIVITY ---
Mode: Sightseeing
Entropy: 1.478
Gini Index: 0.688
Imbalance Ratio: 479.0
Class Proportions (%):
activity
Sightseeing          49.95
Other                16.48
Relaxing             15.22
Hiking                9.80
Exploring             3.96
Unknown               2.29
Water Activities      1.25
Adventure             0.94
Food                  0.10
Name: proportion, dtype: float64
```

```
--- MOOD ---
Mode: Excited
Entropy: 1.5
Gini Index: 0.762
Imbalance Ratio: 6.02
Class Proportions (%):
mood
Excited      32.64
Romantic     21.58
Happy        20.65
Curious      19.71
Calm          5.42
Name: proportion, dtype: float64

--- COUNTRY ---
Mode: Italy
Entropy: 3.929
Gini Index: 0.969
Imbalance Ratio: 65.0
Class Proportions (%):
country
Italy          6.78
Japan          5.74
Switzerland    5.21
France         5.11
Spain          5.01
Romantic     21.58
Happy        20.65
Curious      19.71
Calm          5.42
Name: proportion, dtype: float64
```

Figure 2-3 : Descriptive Statistics of the Dataset

Overall, the descriptive analysis shows that although the dataset is rich and geographically diverse, several labels, especially weather, time of day, and activity, suffer from noticeable class imbalance. Dominant categories such as Sunny, Afternoon, and Sightseeing clearly outweigh the rarer classes such as Rainy, which introduces an important challenge for classification. At the same time, moderate to high entropy values for mood, season, and country suggest good

diversity within these attributes, making them valuable contextual features that can help improve the model's ability to generalize.

2.5 Data Visualization

Correlation Matrix of Labels: This plot shows the linear relationships between all categorical attributes, highlighting moderate correlations between all semantic labels.

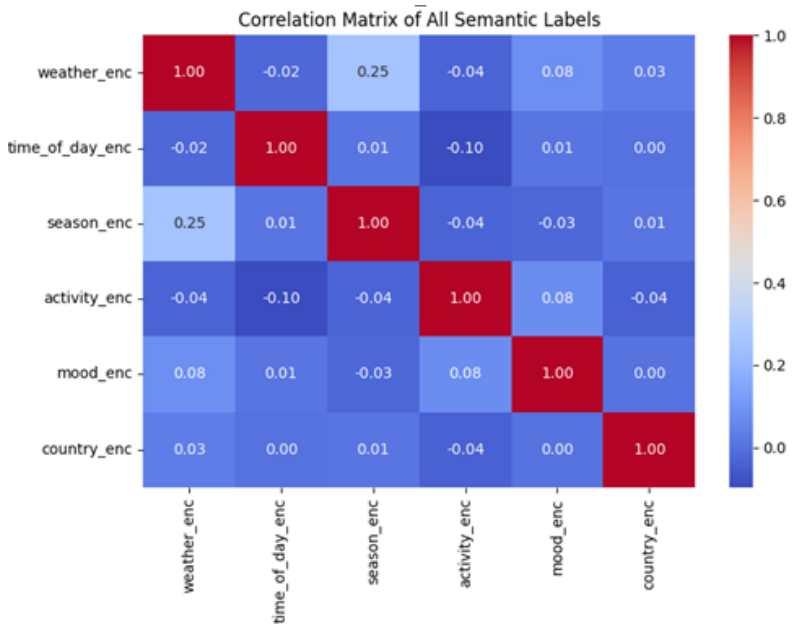

Figure 2-4 : Correlation Matrix

Relationship Visualizations: Stacked bar charts were used to illustrate the relationships between weather and contextual attributes such as season, time of day, activity, mood, and country. These visualizations revealed clear semantic patterns, including the higher occurrence of sunny weather during summer and daytime, as well as the strong association between outdoor activities and clear weather conditions. The plots shows Weather vs Time of Day, and the second one is Weather vs Activity.

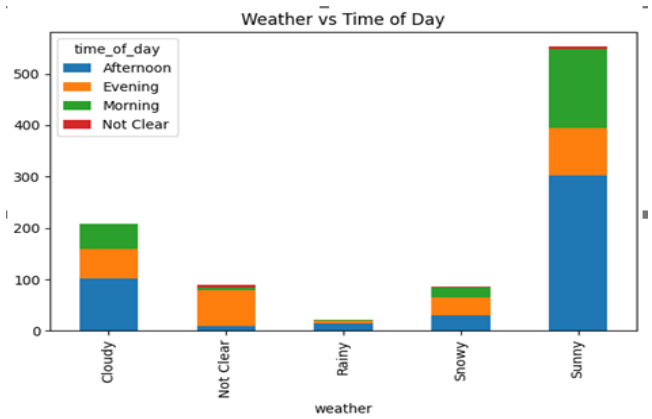

Figure 2-5 : bar charts to illustrate the relationship with time

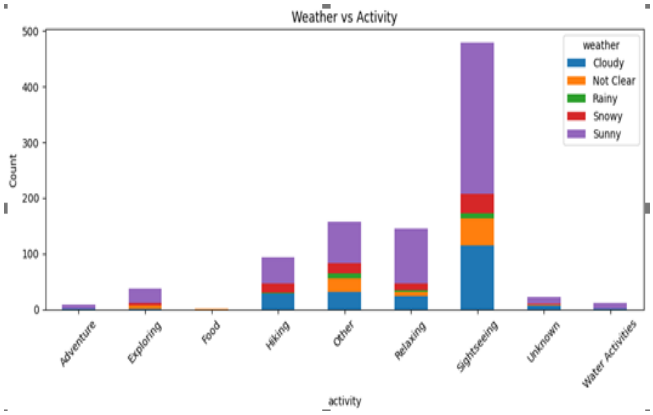

Figure 2-6: bar charts to illustrate the relationship with Activity

2.6 Observed Trends and Dataset Challenges

Key Trends

- Strong Dominance of Sunny Weather: Sunny weather represents more than half of the dataset (57.66%), indicating that most travel images were captured under clear weather conditions.
- Activity–Weather Dependency: Outdoor activities such as sightseeing and hiking occur predominantly under sunny conditions, whereas indoor or passive activities appear more frequently during cloudy or rainy weather.
- Clear Seasonal and Temporal Patterns: Sunny and clear weather conditions appear more frequently in summer and during daytime periods, while cloudy and rainy conditions are

more common in colder seasons. This confirms a strong relationship between weather, season, and time of day.

Key Challenges

- Severe Class Imbalance: Several attributes suffer from strong class imbalance, particularly weather, and activity. Rare classes such as rainy weather and food-related activities are poorly represented, which may negatively impact classification accuracy for these categories.
- Extreme Imbalance in Activity Labels: The activity attribute exhibits the highest imbalance ratio (479), indicating that many activity types have very limited samples, making it difficult for models to learn reliable patterns.

3. Baseline Model (kNN)

As a baseline model, the k-nearest neighbors (kNN) classifier was used with Euclidean distance on the extracted image feature vectors. The purpose of this baseline is to provide a simple reference for comparison with more advanced models. Two values of k were evaluated, namely k=1 and k=3.

3.1 kNN with K = 1

For k=1, the model achieved an overall accuracy of 47.5%. Performance varied across weather classes. The Sunny class achieved the highest recall (0.64) and F1-score (0.66), mainly due to its dominance in the dataset. In contrast, minority classes such as Cloudy showed low precision and recall. The confusion matrix indicates that many samples from other classes were incorrectly predicted as Sunny, highlighting the impact of class imbalance on the model's performance.

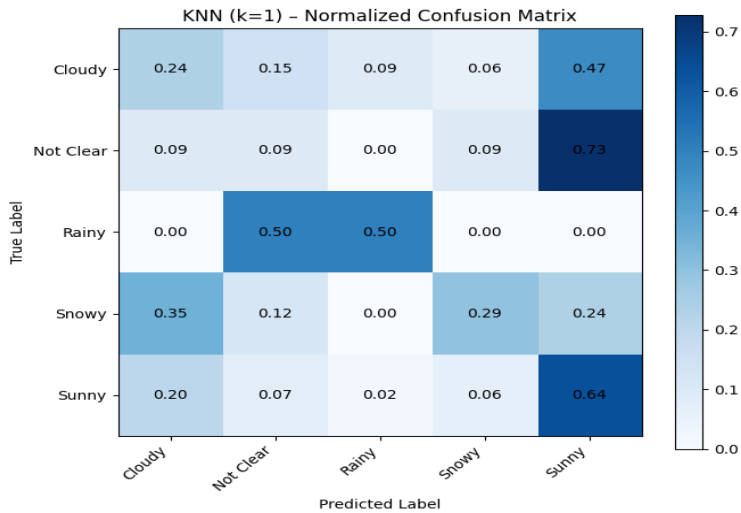

Figure 3-1: confusion matrix of K=1

3.2 kNN with K = 3

Increasing the number of neighbors to k=3 improved the overall accuracy to 53.8%. This improvement is mainly because the model became less sensitive to noise and outliers, leading to better performance for the Cloudy and Sunny classes. The Sunny class achieved a recall of 0.73 and an F1-score of 0.73.

However, the Snowy class was not correctly classified in this setting, resulting in zero recall. This behavior is likely due to the small number of Snowy samples in the dataset and their visual similarity to other weather conditions, which made them difficult for the kNN model to predict

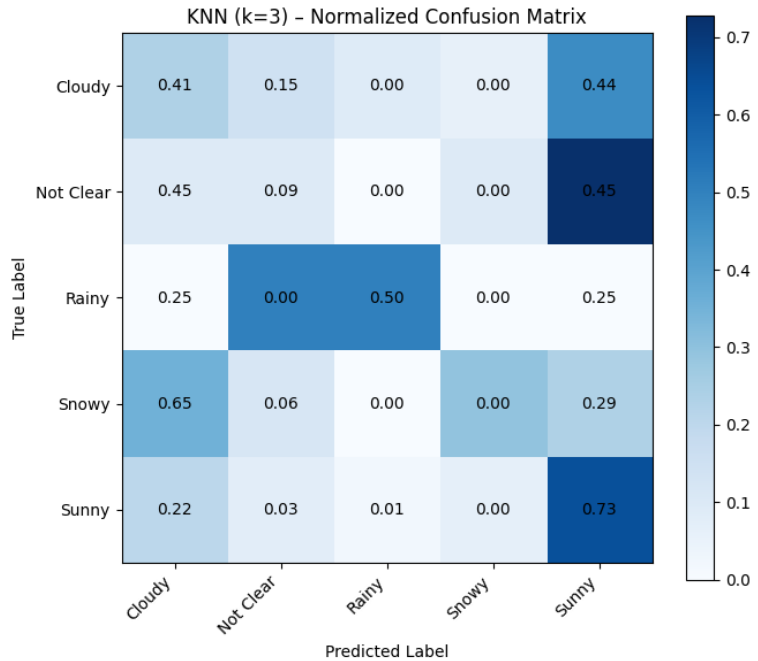

Figure 3-2 : confusion matrix of k=3

3.3 Comparison and Discussion

Overall, (k = 3) outperformed (k = 1) in terms of accuracy and F1-score, suggesting better generalization. However, both models struggled with minority weather classes, as reflected by the low F1-scores for these categories. The confusion matrices reveal a strong bias toward predicting the dominant Sunny class, indicating that kNN performance is heavily influenced by class imbalance and global feature similarity.

K value	Accurcy	F1-score
K=1	0.475	0.33
K=3	0.537	0.344

4. Proposed Machine Learning Models

4.1 Random Forest Classifie

4.1.1 Model Selection and Motivation

This classifier was selected because its ensemble learning method that combains multiple decision trees to improve predictive performance . its well suited for this task because they can handle high-dimensional feature spaces, are relatively robust to overfitting, and are capable of capturing complex non-linear relationships in image features.these make the random foresta strong choice for weather classification based on extracted visual features.

4.1.2 Feature Representation

The model was trained using image features extracted from the travel images. These features include HSV color histograms and basic statistical descriptors, which capture global color distributions and overall image characteristics relevant to different weather conditions

```
...   Feature matrix shape: (800, 103)
      Number of features per image: 103
```

Figure 4-1: Feature Representation

4.1.3 Hyperparameter Tuning

To optimize performance , several Random forest parameters were tuned using the validation set , specifically , the following parameters

4.1.3.1 max_depth

The max_depth parameter was evaluated using values [5,10,20,30]. The highest validation accuracy (66.7%) was achieved with max_depth = 10.

```
Tuning max_depth...
  max_depth=5 -> Weather: 0.6583333333333333
  max_depth=10 -> Weather: 0.6666666666666666
  max_depth=20 -> Weather: 0.65
  max_depth=30 -> Weather: 0.6416666666666667
```

Figure 4-2: max_depth parameters

The results indicate that a moderate tree depth provides the best generalization for this dataset. When the maximum depth was too small (max_depth=5), the model underfit the data and failed to capture important variations in weather-related visual features. In contrast, deeper trees (max_depth=20 and 30) slightly reduced validation accuracy, suggesting overfitting. Therefore, max_depth = 10 achieves a good balance between model complexity and generalization.

4.1.3.2 min_samples_leaf

the min_samples_leaf parameter was evaluated with values [1,2,4,8]. The best validation accuracy (66.7%) was achieved when min_samples_leaf = 1 or 2, while larger values led to a gradual decrease in performance. Increasing the minimum number of samples per leaf restricts tree growth and can reduce overfitting; however, in this case, higher values caused the model to underfit and reduced its ability to capture important patterns in the data. Therefore, min_samples_leaf = 1 was selected.

```
Tuning min_samples_leaf...
  min_samples_leaf=1 -> Weather: 0.6667
  min_samples_leaf=2 -> Weather: 0.6667
  min_samples_leaf=4 -> Weather: 0.6500
  min_samples_leaf=8 -> Weather: 0.6250
```

Figure 4-3: min_samples_leaf parameters

The min_samples_leaf parameter controls how detailed each tree leaf can be. In this dataset, increasing its value reduced validation accuracy, indicating that the model benefits from learning finer patterns in the image features. Therefore, min_samples_leaf = 1 was selected as it achieved the best performance.

4.1.4 Performance Evaluation

After tuning, the model was evaluated on the test set using accuracy as the primary metric, along with precision, recall, and F1-score. A confusion matrix was also used to analyze misclassification patterns.

The model achieved an accuracy of 66.7% and a macro F1-score of 0.51, showing a clear improvement compared to the baseline model. The class-wise results indicate that the Sunny class achieved the highest performance, with a recall of 0.93 and an F1-score of 0.78, reflecting its dominance in the dataset. The Not Clear and Rainy classes achieved moderate F1-scores of 0.60 and 0.50, respectively, despite their smaller sample sizes. In contrast, the Cloudy and Snowy classes showed lower recall values, indicating that these weather conditions remain challenging for the model to distinguish based on the extracted visual features alone.

Class	accurcy	samples	errors
cloudy	0.17	23	19
snowy	0.31	13	9
Rainy	0.33	3	2
Sunny	0.93	70	5
Not clear	0.54	11	5

The normalized confusion matrix reveals that many misclassifications occur when Cloudy, Snowy, and Rainy images are predicted as Sunny. This pattern highlights the impact of class imbalance and the visual similarity between certain weather conditions under bright lighting.

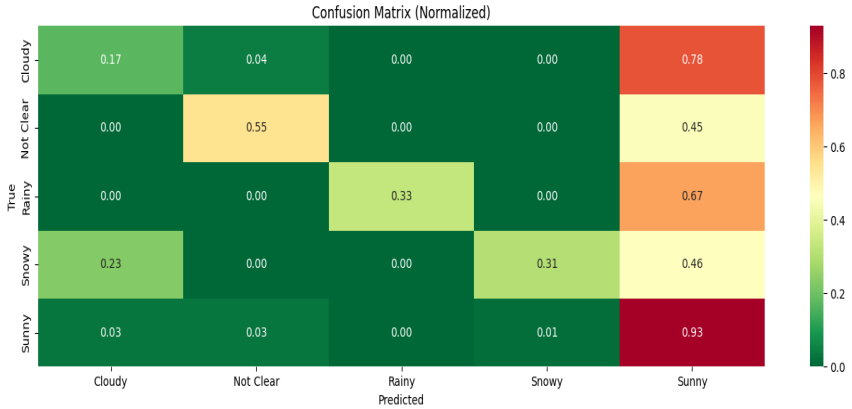

Figure 4-4 : confusion matrix of Random forest

4.1.5 Comparison with Baseline

Compared to the kNN baseline, which achieved accuracies of 47.5% for k=1 and 53.8% for k=3, the Random Forest model showed a clear improvement in overall performance. This improvement is mainly due to Random Forest's ability to learn non-linear decision boundaries and capture interactions between features, while kNN relies only on distance-based similarity. However, despite this improvement, the model still struggles with minority classes, which is largely caused by class imbalance and the visual similarity between different weather conditions.

4.2 Support Vector Machine (SVM) Classifier

4.2.1 Proposed Models and Feature Representation

To predict weather conditions from travel images, two complementary visual feature extraction techniques were employed to each image, that capture both color information and local image details.

- HSV Color Histogram Features

Each image was first resized to 224 × 224 pixels and converted into the HSV color space. Then, for each of the Hue, Saturation, and Value channels, a 16-bin histogram was extracted and combined to form a 48-dimensional feature vector for every image. These features capture color distribution and illumination characteristics which are especially helpful for distinguishing between weather conditions such as sunny, cloudy, rainy and snowy scenes.

- SIFT Bag-of-Visual-Words (BoW) Features

SIFT descriptors were extracted from grayscale versions of the images and grouped using MiniBatch K-Means with k = 200 to build a visual vocabulary. Each image was then represented as a 200 dimensional Bag-of-Words histogram that summarizes its texture and structural patterns that would be helpful to classify weather conditions such as rainy and snowy scenes.

- Final feature vector

The HSV and SIFT BoW vectors were concatenated to form a 248-dimensional final feature vector, combining color, illumination, texture, and structural information for robust weather representation.

4.2.2 SVM model

The Support Vector Machine (SVM) model was selected because of its strong performance in high dimensional image classification tasks and its ability to handle non-linear decision boundaries through the use of the RBF kernel. SVM is particularly effective when the number of features is large relative to the number of samples, making it well suited for the combined HSV and SIFT feature representations. Additionally, its robustness to noise and class overlap makes it an appropriate choice for visual weather prediction.

The dataset was split using stratified sampling into 60% for training, 20% for validation, and 20% for testing, the training set was used to learn the SVM model parameters, while the validation set was dedicated exclusively to hyperparameter tuning, ensuring that the original class distribution was preserved across all subsets. All feature vectors were then standardized using a StandardScaler so that each feature contributes equally during learning

4.2.3 Hyperparameter Tuning

The SVM regularization parameter C was tuned using a logarithmic grid: C = [0.01, 0.1, 1, 10], each of the regularization parameters C were evaluated using validation set in order to find the optimal balance between maximizing the margin and minimizing classification error. The value producing the highest validation accuracy was selected as the optimal parameter. The optimal C that was selected C=1 due to the high Validation Accuracy = 0.5929. Although C = 1 and C=10 achieved the same validation accuracy, C=1 was selected because it produced a higher final test accuracy, indicating better generalization performance.

```
C = 0.01  → Validation Accuracy = 0.5786
C = 0.1  → Validation Accuracy = 0.5786
C = 1  → Validation Accuracy = 0.5929
C = 10  → Validation Accuracy = 0.5929
Best C based on validation: 1
```

Figure 4-5: SVM regularization parameter

Then the model was trained on the merged training and validation sets using the best hyperparameter, to learn from a larger amount of data. To handle class imbalance detected in EDA, the final SVM model was trained using the,and class_weight = balanced, giving more importance to rare weather classes.

4.2.4 Performance Evaluation

After test the model on unseen data, the model performance was evaluated using:

- Overall Accuracy = 0.56428
- Precision, Recall, and F1-Score per class

```
Final weather test Accuracy: 0.5642857142857143

Weather Classification Report:

              precision    recall  f1-score   support

      Cloudy       0.32      0.40      0.36        30
   Not Clear       0.42      0.38      0.40        13
       Rainy       0.00      0.00      0.00         3
       Snowy       0.67      0.29      0.40        14
       Sunny       0.68      0.72      0.70        80

    accuracy                           0.56       140
   macro avg       0.42      0.36      0.37       140
weighted avg       0.56      0.56      0.56       140
```

Figure 4-6 : SVM Performance Evaluation

The SVM model achieved an overall test accuracy of 56.4%, which is a reasonable result considering the strong class imbalance in the dataset. The model performs best on the dominant Sunny class, with an F1-score of 0.70, showing that the combined HSV and SIFT features are effective in capturing clear-weather visual patterns.The Cloudy, Not Clear, and Snowy classes

show moderate performance, with F1-scores between 0.36 and 0.40. However, the Rainy class was not predicted correctly due to its very small number of samples, clearly illustrating the negative impact of severe class imbalance. The low macro averaged F1-score (0.37) further confirms that the model struggles with rare weather categories.

- Confusion Matrix visualization

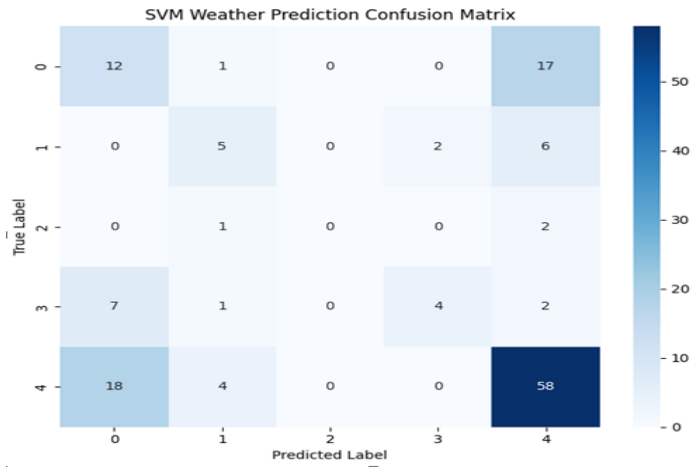

Figure 4-7: SVM Confusion Matrix

Cloudy = 0, Not Clear = 1, Rainy = 2, Snowy = 3, Sunny = 4

The confusion matrix shows that the model correctly predicts most Sunny samples, while frequent confusion occurs between Cloudy, Not Clear, and Snowy classes. The Rainy class is not correctly recognized due to its very small number of samples. This confirms that class imbalance significantly affects the classifier's performance on minority weather conditions.

4.3 Model Comparison and Selection

Both the Random Forest and SVM models improved upon the kNN baseline; however, their performance differed. The Random Forest model achieved the highest test accuracy (66.7%) and a higher macro F1-score (0.51), showing better overall performance and generalization. It handled most weather classes more consistently, although it still struggled with minority classes due to class imbalance.

The SVM model achieved a lower test accuracy (56.4%) and macro F1-score (0.37). While it performed well on the dominant *Sunny* class, it had difficulty predicting rare classes, especially Rainy, despite using class weighting.

Overall, the Random Forest model was selected as the best model for this task because it achieved higher accuracy and more balanced performance across weather classes compared to SVM

5. Performance analysis

5.1 Error Analysis Methodology

To better understand how the best-performing model (Random Forest) behaves, a detailed error analysis was carried out on the test set predictions. This analysis focused on examining misclassified samples and identifying recurring patterns in the model's errors. Per-class accuracy was calculated using the confusion matrix to determine which weather categories were predicted reliably and which were more challenging. Misclassification frequencies were then measured to highlight the most common confusion trends between weather classes. In addition, the class distribution was analyzed to evaluate the impact of class imbalance on the model's performance.

5.2 Observations and Insights

5.2.1 Class Specific Performance

The model demonstrates very strong performance on the Sunny class, achieving an accuracy of 92.86%, which confirms that clear weather scenes are visually distinctive and are learned well by the classifier. In contrast, Cloudy is the weakest performing class with an accuracy of only 17.39%, followed by Snowy (30.77%) and Rainy (33.33%). These results indicate that overcast and adverse weather conditions are considerably more difficult for the model to recognize reliably.

5.2.2 Effect of Class Imbalance

A major factor contributing to these misclassifications is the severe class imbalance ratio of 23.3:1. Sunny scenes dominate the dataset, while the Rainy, Snowy, and Cloudy classes contain

far fewer samples. This imbalance causes the model to become strongly biased toward predicting the Sunny class, even in cases where visual features suggest a different weather condition.

5.2.3 Confusion Patterns

The most common misclassification pattern is Cloudy → Sunny, which occurs 18 times and accounts for 78.3% of all Cloudy samples. This suggests that cloudy scenes often share visual similarities with sunny ones, such as overall brightness and color composition, making them difficult for the model to distinguish.

Other frequent confusion patterns include Snowy → Sunny (46.2%), Not Clear → Sunny (45.5%), and Rainy → Sunny (66.7%). These results further confirm that, when uncertainty arises, the model tends to default to predicting the Sunny class.

5.2.4 Data-Related Challenges

These error patterns point to several possible data-related challenges, including:

- A limited number of training samples for minority weather classes.
- Visual overlap between different weather conditions (for example, bright cloudy skies that resemble sunny scenes).
- Limited diversity within rare classes, which reduces the model's ability to generalize effectively.

5.3 Visualization Performance

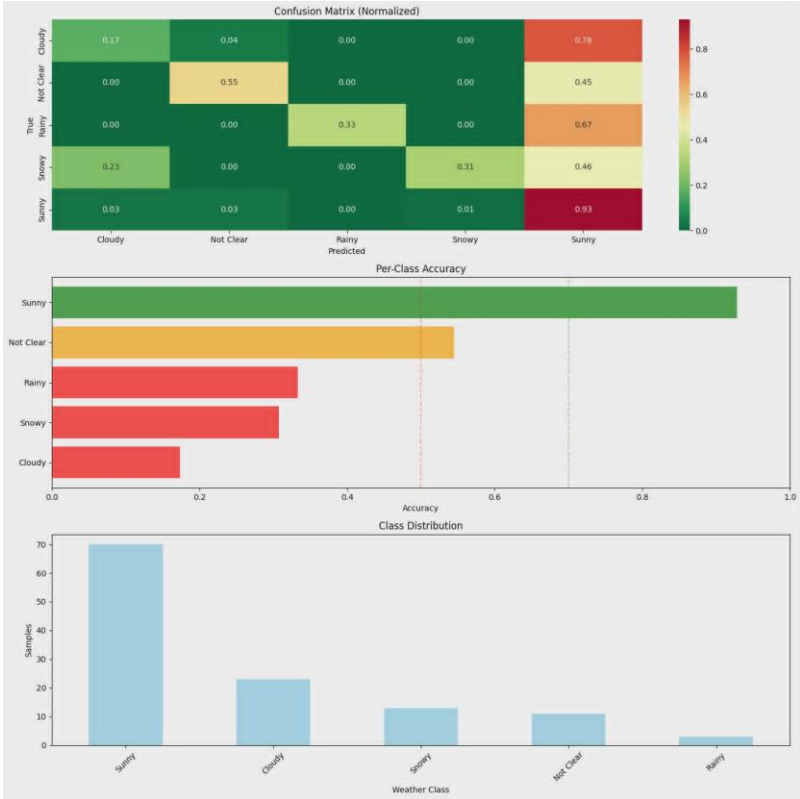

Figure 5-1: Visualization Performance of best model

The normalized confusion matrix shows that most misclassifications happen when Cloudy, Snowy, Not Clear, and Rainy samples are incorrectly predicted as Sunny, revealing a strong bias toward the dominant class. The per-class accuracy plot further highlights this pattern, with Sunny achieving the highest accuracy while Cloudy and Snowy show the lowest, making them the most challenging categories for the model to classify. The class distribution plot helps explain this behavior by illustrating the severe imbalance in the dataset—Sunny contains far more samples than the other weather classes, especially Rainy and Snowy, which are highly underrepresented.

6. Conclusions and Discussion

6.1 Main Findings

In this project , several machine learning models were developed and evaluated to predict weather conditions from travel images using extracted visual features. A kNN classifier was first implemented as a baseline, followed by two more advanced models: Random Forest and Support Vector Machine (SVM). The results demonstrate that model choice and feature representation have a significant impact on classification performance.

The kNN baseline achieved accuracies of 47.5% for k=1 and 53.8% for k=3, showing limited performance and a strong bias toward the dominant Sunny class. The Random Forest model achieved the best overall performance, reaching an accuracy of 65% and a macro F1-score of 0.50, indicating improved generalization and better handling of non-linear relationships in the feature space. The SVM model achieved a moderate accuracy of 56.4%, performing well on the Sunny class (F1-score of 0.70) but struggling with minority classes such as Rainy due to limited samples.

Across all models, performance was consistently higher for the Sunny class, reflecting its dominance in the dataset. In contrast, weather conditions such as Cloudy, Snowy, and especially Rainy were more difficult to predict. The confusion matrices revealed frequent misclassifications of minority classes as Sunny, highlighting the impact of class imbalance and the visual similarity between different weather conditions.

6.2 Limitations

Throughout this project, several limitations were observed that may influence the overall performance and generalization of the proposed system. The dataset shows a strong class imbalance, especially in the weather and activity categories. Rare classes such as *Rainy*, *Snowy*, and several activity types are underrepresented, which makes it difficult for the model to learn reliable patterns for these categories and causes it to favor dominant classes during prediction.

In addition, the system mainly depends on handcrafted visual features (HSV and SIFT), which may not fully capture the rich and complex semantic information present in real-world scenes.

This limitation can make it harder for the model to distinguish between visually similar conditions, such as cloudy and sunny weather.

Finally, using overall accuracy as the main evaluation metric can be misleading when working with imbalanced datasets.

6.3 Possible Improvements and Future Directions

Further improvements could include experimenting with ensemble learning methods and more advanced feature extraction techniques to better handle complex visual variations between weather categories. Applying class-balanced loss functions and resampling strategies may improve recognition of underrepresented weather conditions. In addition, exploring and comparing multiple classification models such as deep learning-based CNN models would help identify the most suitable architecture for visual weather recognition and potentially improve overall prediction performance.